

**LETTERKENNY MUNITIONS CENTER PURCHASE DESCRIPTION FOR
AFT SEALANT FD4120AS/FD4120BS & FWD SEALANT UR4515AG/UR4515BG**

1. REQUIREMENTS: Letterkenny Munitions Center has the requirement for AFT SEALANT FD4120AS/FD4120BS & FWD SEALANT UR4515AG/UR4515BG. This will potentially be a base plus 4 option year contract. with the following standard features and options:

1.1 SPECIFICATIONS:

1.1.1 Must be two-part, fast curing, urethane adhesive, effective in minus 54 to plus 94 degrees Celsius.

1.1.2 MIL-STD-129 Marking for shipping and storage

1.1.3 MIL-STD-186 Protective Finishing for Army Missile Systems

1.1.4 Must adhere to American Society for Testing and Materials – ASTM D 1002 – Test for strength properties of adhesives in shear by tension loading

1.1.5 Must adhere to American Society for Testing and Materials – ASTM D 2651 – Preparation of metal surfaces for adhesive bonding recommended practices

1.1.6 Must be a material capable of being mixed and extruded by a suitable two-feed line extrusion unit for a high production rate and surface application of an adhesive bead. The adhesive bead shall retain adhesive characteristics as defined in 1.1.1 to 1.1.5.

1.1.7 The Part A and Part B components shall be packaged in separate containers – Part A is the base, urethane adhesive and Part B is the accelerator, urethane adhesive.

1.1.8 The application pot life of the two-part formulated product after mixing shall be 15 plus or minus 5 minutes at room temperatures between 21 and 24 degrees Celsius and at relative humidity up to 75%.

1.1.9 The Part A and Part B components in liquid form shall mix readily to a consistency suitable for application. After Part A and Part B mixing surface application, the fast curing adhesive bead (ribbon) shall be capable of being applied to horizontal and vertical surfaces without runoff or sag. Sag shall be no greater than 1.6 millimeters (mm) at 27 degrees Celsius and 3 mm at 70 degrees Celsius.

1.1.10 After developing full required strength in seven days at room temperature, 25 degrees Celsius, the adhesive product shall have a minimum lap shear strength of (1) bond strength of minus 40 degrees Celsius then a 1.4 minimum average lap shear strength, (2) bond strength of 25 degrees Celsius then a 5.5 minimum average lap shear strength and (3) bond strength of 94 degrees Celsius then a 1.7 minimum average lap shear strength.

1.1.11 After exposure to 94 degrees Celsius for a period of three hours, the minimum solid content by weight of Part A or B product components shall be no less than 93 percent. Following curing, the adhesive shall be greater than 97 percent solids.

1.1.12 The Part A and Part B components shall be extrudable using a Graco Hydro-cat Proportioning and Dispensing Machine with a fluid to air pressure ratio of 34 to 1 or equivalent equipment, which shall mix the two components in the head and extrude controlled end product flow through head applicator.

1.1.13 The Part A and Part B adhesive components shall meet all requirements defined herein when stored in unopened original containers at temperatures between plus 4 and 38 degrees Celsius for a minimum storage life of nine months from the date of manufacture. The storage life may be extended an additional six months for a total of 15 months by recertification of the characteristics listed in 1.1.8 through 1.1.12. The storage life may be extended at three-month intervals thereafter through recertification of the characteristics in 1.1.8 through 1.1.12.

1.1.14 The Part A and Part B adhesive components shall not separate or exhibit any settling out of ingredients throughout its usable life.

1.1.15 Lot size shall consist of all the Part A and Part B material manufactured at one time, from one batch, forming part of one contract r order, and submitted for acceptance at one time and place.

1.1.16 Marking: Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and shall include the following:

- 1.1.16.1 Adhesive (Part A or B) two-part
- 1.1.16.2 Specification number and revision
- 1.1.16.3 Manufacturer's material nomenclature
- 1.1.16.4 Name and address of manufacturer
- 1.1.16.5 Quantity of material or net weight
- 1.1.16.6 Date of manufacture and batch or lot number
- 1.1.16.7 Mixing instructions
- 1.1.16.8 Deterioration and storage life instructions
- 1.1.16.9 Contract or purchase order number

1.1.17 Must meet intended use: This compound is intended for use as an adhesive for bonding components in hazardous areas to be subjected to service temperatures minus 54 to plus 94 degrees Celsius.

2. **WARRANTY:** A standard commercial warranty shall be furnished with each separate unit.

3. **PRESERVATION, PACKAGING AND DELIVERY:** The contractor shall utilize standard commercial methods for preservation and packaging appropriate for the AFT SEALANT FD4120AS/FD4120BS & FWD SEALANT UR4515AG/UR4515BG. As a minimum, all areas susceptible to damage from exposure to the elements shall be preserved and/or packed to prevent damage. The contractor shall be responsible for ensuring the equipment is delivered to Letterkenny Munitions Center, Building 3315 in good condition.

All deliveries shall be made between the hours of 7:00 AM and 3:00 PM, Monday through Thursday with the exclusive of Federal Holidays.

4. **SECURITY, SAFETY, AND FIRE PREVENTION:** The contractor shall comply with LEAD P 385-4, Requirements for Security, Safety, and Fire Prevention for Contractors Performing work on Letterkenny Army Depot.
5. **FINAL ACCEPTANCE:** Final acceptance shall be contingent upon successful completion of all requirements of this purchase description.
6. **POINT OF CONTACT:** Will be determined at time of award.